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Original Article

Can the Aviation Industry be Useful in Teaching Oncology about Safety?

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Abstract

Healthcare practitioners have long considered aviation as a domain from which much can be learned about safety. Over the past 30 years, attempts have been made to apply aviation safety-related concepts to healthcare. Although some applications have been successful, a few decades later, many healthcare safety experts have learned that the appeal of the aviation–healthcare analogy is an illusion. Both domains are so basically dissimilar that simple adoption of aviation concepts will not be successful. However, what has succeeded is healthcare's adaptation of specific aviation safety concepts. Three concepts, investment in safety, human factors and safety management systems, are described and examples are given of adapted applications to healthcare/clinical oncology. Finally, there is a need to ensure that these concepts are applied systematically throughout healthcare rather than sporadically and without a centralised mandate, to help ensure success and improved patient and provider safety.

Key words: Anaesthetics; Aviation; healthcare: oncology; human factors; safety; safety management systems

Introduction

> If you are looking for perfect safety, you will do well to sit on the fence and watch the birds (Wilbur Wright, 1901).

Radiation oncology, like other specialties in healthcare, considers the safety of its patients and personnel to be extremely important. Given the impressive improvements in safety in the aviation industry, it is logical for oncologists to look at those successes as a possible source of safety improvement initiatives. Aviation safety can be considered to have started in 1783, when the Montgolfier brothers launched a balloon to 1500 feet carrying a sheep (with human-like physiology), a duck (accustomed to flying at altitude) and a rooster (which was not) [1]. By substituting animals for humans, the Montgolfiers improved safety, at least for humans. This was also an example of one of the first uses of human factors in aeronautics – by approximating human physiology with a sheep. Another, less successful, example is that of a 1908 flight captained by Orville Wright, with Army Lieutenant Selfridge as an observer. A propeller blade broke, severing a wire to the rudder: the plane nose-dived to the ground from about 75 feet. Wright was badly injured. Selfridge died despite immediate neurosurgery. However, this first fatal accident produced direct and rapid improvements to aviation safety, including changes to the aircraft.

In 1972, the Secretary General of the Canadian Medical Association wrote of hearing a speaker describe the need for 'limited' licensure and 'regular re-assessment of competence of the procedures carried out to maintain high standards of performance'; he considered the comparison to be

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'very valid' [2]. The speaker, however, was not talking about doctors but pilots.

A decade later, Dr R.B. Lee, an Australian aviation safety expert, spoke at an Australian anaesthetic meeting about the analogy between aviation and anaesthetics. He compared pilots with anaesthetists and the aviation system with the anaesthetic system. He emphasised the importance of human performance in the context of the environment and associated equipment, the 'man–machine interface', for both pilots and anaesthetists. He foreshadowed future understanding of the critical importance of the role of organisations and regulatory agencies in safety, stating that corporate factors represented the 'area of greatest potential for the enhancement of safety in aviation'. He considered the situation no different in anesthetics, giving the example of a patient harmed by multiple factors of 'less than optimum personnel, equipment, and working conditions, because of hospital policy, which is in turn determined by government ('the corporation')' [3].

At the same time, anaesthetists elsewhere were similarly engaged in learning about aviation safety. For example, Dr David Gaba of Stanford University developed the first modern full-body patient simulator and then introduced aviation crew resource management training to healthcare [4–6]. By the early 1990s there seemed to be little objection to the wholesale importation of aviation safety principles and practices into anaesthesia and medicine in general. Although programmes such as Gaba's and others have been successful, two decades later, many experts have learned that the aviation–healthcare analogy, although appealing, is illusionary. Aviation and healthcare are dissimilar in their basic constructs, and to simply adopt aviation's ways of doing things can lead to eventual failure.

What has succeeded, however, is the adaptation by healthcare of specific concepts of aviation safety. For example, one of the authors (JMD) previously collaborated with an aviation safety expert to adapt air accident investigation techniques to healthcare. The result was a systematic, human factors-based methodology specifically designed for the investigation of anaesthetic-related deaths [7]. This systematic systems analysis [8] has been expanded and refined, with the investigation's orientation on the system and not the individual(s) involved, and applied to all types of adverse outcome [9], from single cases [10] to large, multiple victim inquests [11].

However, when adapting specific concepts from aviation, 'lessons learned and illustrative materials' should come from healthcare and not aviation [12]. Although it might be fascinating to watch video clips of real or reconstructed catastrophes from aviation (and other industries), as Hunt and Callaghan [13] stated, 'it is too simplistic to equate the performances of aviation teams that operate and maintain aircraft with surgical teams delivering healthcare'.

Safety Concepts from Aviation

There are at least three important aviation safety concepts that can (and should) be applied to healthcare, although their applications to date must be criticised for the lack of a system-wide approach. These concepts are: (i) investment in safety; (ii) human factors; and (iii) safety management systems (SMS).

Investment in Safety

Healthcare providers are all too aware of financial restraints, with budget restrictions leading to cutbacks in staffing, equipment and even housekeeping. By contrast, aviation has intuitively realised that it is more economical to be 'safe' than to investigate the accident and deal with its aftermath, although aviation has not always considered that costs [14] are really an investment in safety. To illustrate this point, Lercel and colleagues [15] developed a three-level model, giving examples of an airline's stock value after an accident, the positive and negative returns on investment in safety programmes and the costs and benefits of a particular safety intervention.

Application to Healthcare/Clinical Oncology

Although many healthcare regulatory authorities and organisations have calculated the costs of harm to patients, this has been partly to minimise the organisation's liability rather than maximising safety. Often these calculations are reactive to patients being harmed rather than proactive to threats of harm. Making this type of decision relates to an organisation's culture, values and priorities. Consequently, few medical departments have staff trained specifically, nor do jurisdictions mandate specifically skilled staff or specifically mandated clinical positions, in human factors and techniques of systems analysis, including failure mode and effects analysis (FMEA).

Most published oncology incident reports are either retrospective reviews of event rates or prospective evaluations of department-derived safety or incident reporting systems. These reports lack both positive and negative financial assessments of the potential effects of implementing safety measures, including potential productivity changes from system re-engineering. For example, when staffing changes are recommended, they are often in response to *post-hoc* incident-review findings, rather than in reaction to proactive processes identifying hazards.

Human Factors

The concept of human factors has been an intrinsic part of safety thinking in aviation since World War II, with development of the 'man–machine model'. Integrating the abilities of both man and machine helped reduce the death toll from designing and manufacturing aircraft without considering human information processing limitations [16]. Modelling this interaction continued, with development of the SHEL model: software (rules, policies), hardware (equipment), environment and liveware (humans). Hawkins [17] expanded the model to include liveware–liveware interaction (the SHELL model). The 'science of human factors' then started to contribute to the ever-increasing safety of aviation, although with some departure from an integrated model, as concepts such as ergonomics, cognition and perception developed and came to be seen as discrete fields. In reality, human factors concerns all aspects of a human's abilities and limitations to process information in one form or another [3].

Application to Healthcare/Clinical Oncology

In healthcare, the application of human factors has been fragmented, often focusing only on problems such as fatigue, workload [18,19] and communication. This approach seems to represent more of an application of 'the human factor', looking only at the personnel (and often one person) involved in adverse events. Indeed, Lee [3] cautioned against a narrow approach to human factors, suggesting that it would be 'unwise for anesthesiologists to adopt only one element of aviation human factors, such as equipment ergonomics'.

Healthcare needs to adopt an approach in which the view of the human is integrated into the environment, and working under the canopy of the organisational and the regulatory agencies, thus providing the context of the work and human–human relationships. Ideally, this approach should be driven by some kind of model, helping ensure that all contributing parts of the model are considered when reviewing safety [11,20].

For example, using the man–machine model and human factors methods, Chan *et al.* [21] reported a department-wide approach to identify human factors issues in radiation therapy delivery processes. The researchers carried out an analysis of workflow, observing radiotherapy students using the system and looking for areas with which they had a high probability of problems. The researchers then iteratively redesigned the system, based on the initial results and from retesting the redesigns. These changes in the structure of the delivery system contributed to changes in the process of care (faster task completion, reduction in use errors [22]) and in staff outcome (greater satisfaction with the new system) [21]. However, departments elsewhere (may) lack staff with formal training in these methodologies, leading to such studies being rare and their findings, such as the structural changes, not widely implemented.

Another area in which human factors methodologies can be applied is that of equipment and processes that require human transfer of data, with the intrinsic probability of transcription errors. Changes in the system's structure, modernising from manual to computerised processes with automatic data transfer, will help reduce transcription errors, although they will not prevent or eliminate errors. However, as in all activities, human limitations will be exposed if staff members do not understand the complexities of data being transferred. There are also the well-recognised problems of failing to anticipate that errors could be made or believing that all computerised data are correct [23]. This is of particular concern in radiotherapy with the use of data when commissioning linear accelerators and planning systems. Erroneous data can be considered a latent condition [24] in the system that may take years to detect and could result in a large number of serious adverse patient outcomes, including death [25]. What is required is equipment that is designed, manufactured and maintained to be significantly robust to anticipate what erroneous data could look like and to have ongoing monitoring systems to ensure such problems are minimised. Clinicians need to work closely with equipment vendors to design such systems.

Development of a Comprehensive Safety Management System

An SMS is a systematic and structured approach to managing safety, through specific organisational structural elements, including governance, policies and procedures, hazard identification, safety performance measurement and monitoring, and safety communication [26]. This concept was started in the 1950s–1960s by the US Air Force Ballistic Missile Division, with the basic idea that all operations, individuals, pieces of equipment and working environments are part of a system. The 1960s–1970s saw the application of quality management systems, with structured and documented work, and a focus on ensuring product quality while increasing productivity and decreasing costs. In aviation, further safety improvements were based on reactively investigating aircraft incidents and accidents and making changes to avoid the same thing happening again. However, in the 1980s, aviation regulators started to require that quality management systems include a programme for accident avoidance, that is, a proactive approach to safety. This meant looking for factors that could lead to 'trouble' and dealing with them - before 'trouble' occurred. Such actions involved identifying hazards and reviewing close calls ('near hits'), which offer rich lessons about the antecedents of accidents, without the catastrophe of one.

In 1990, Degani and Wiener [27] published their seminal work on a human factors analysis of the use and misuse of, and design guidelines for, the 'normal checklist' used by cockpit crews in their normal (non-emergency) activities. Central to this work was the 'philosophy of use' of the checklist, determined by regulatory agencies, airlines and aircraft manufacturers [28]. They developed an outline for the design of flight deck procedures, based on the three Ps of philosophy, policies and procedures. They suggested flight crews would deviate from standard operating procedures in part because of a 'failure to develop an overall philosophy and set of policies that are consistent with each other, and procedures that are consistent with both' [29]. Later they added 'practices', representing every correct or incorrect action carried out by the flight crew [30]. They reiterated the importance of ensuring what flight crews did, and how, could only be understood if looked at in terms of how standard operating procedures were 'developed, taught and used' [28].

In 2002, Transport Canada had the beginnings of a 'four Ps' SMS, which was a 'new way to address its oversight responsibilities and to improve how Canadian airlines managed safety' [31]. This model was then reduced to three parts. Part 1: 'organisation', required a 'designated "accountable executive"', most often the CEO, responsible for safety, and SMS programme, policies and procedures documentation. Part 2: 'risk management', included reporting, analysing and generating recommendations to mitigate hazards, with employees involved in all safety activities, especially reporting hazards before an accident. Part 3 included gathering safety-related information, verifying effective mitigation of previously identified hazards, and sharing safety information within the organisation and externally [32].

In 2005, the International Civil Aviation Organization (ICAO) combined SMS information from various member states (countries) and organisations into a safety management manual to facilitate a global approach to SMS implementation. By 2006, the ICAO required 'aircraft operators, aerodrome operators, air traffic services providers and maintenance organizations worldwide' to implement an SMS [33]. The model was again defined by four pillars: policy, risk management, safety assurance and safety promotion [34]. The latter represented communication about safety hazards and ensuring that the organisation had what James Reason has described as a 'just culture' [24].

Application to Healthcare/Clinical Oncology

Every healthcare organisation needs an SMS – in some form. The seven decade long history of aviation SMS serves to show the long timeline and multiple sources in its evolution. Healthcare has an advantage in being able to see what aviation has done and to adopt some adapted version of an SMS. Table 1 offers an outline of what such an SMS might comprise.

Table 1 provides the basis by which every department head and hospital CEO can lead and direct safety-related activities, and encourage the development of a safety-oriented culture. The latter starts with the attitudes, actions and behaviours of the leaders. Indeed, managing safety should be part of the job description requirements of every department head to fulfill and every hospital's CEO to endorse and support.

Managing safety requires information and data, and the basis of any SMS is the safety information it provides. However, what might not be evident from the description of an aviation SMS is that policies, procedures and terms are clearly delineated. Having an effective SMS requires that everyone speaks the same language. In fact, an SMS should start with a dictionary, as exemplified by the ICAO's safety management manual [35]. By contrast, the oncology literature provides examples of multiple, varying definitions of certain terms, including 'errors' [18,19,23,36–41], with 'error and incident terminology … often (being) mixed between reports and studies' [41].

A major contributor to this difficulty lies in the International Atomic Energy Agency (IAEA) Safety Standards' definition of 'accident': 'Any unintended event, including operating errors, equipment failures or other mishaps, the consequences or potential consequences of which are not negligible from the point of view of protection or safety' [42]. Contrast this with part of ICAO's definition of an

Table 1
The six Ps of a healthcare safety management system (adapted from [29,30])

Structure	Philosophy • There will always be threats to safety, which are addressed both proactively and reactively. • The organisation, starting with the CEO and the Board, sets the safety standard. • Safety is everyone's responsibility, through developing and maintaining the necessary knowledge, skills and attitudes. Policy • Organisational structures and processes systematically incorporate safety goals into every aspect of the operation. • Safety-critical policies include those for proactively analysing the system and its components, monitoring hazard and incident reports from safety reporting databases managing and disclosing harm, investigating reactively, just culture and communicating internally and externally. • There are clear statements about responsibility, authority and accountability, including not punishing errors.
Process	Procedures • There are clear directions for all, which ensure a lack of conflict with other procedures. • Monitoring, evaluating and managing safety status is ongoing proactively and reactively. • Feedback provides an evolutionary approach to the review of activities. Practices • Where possible, everyone follows well-designed, effective procedures and avoids shortcuts that can detract from safety. • When compliance with procedures is not possible, individuals freely report their difficulties and, together with managers, seek to identify the problems. • Everyone takes appropriate action when a safety concern is identified.
Outcome	Products • There are well-defined, fact-based measures of outcomes. • Outcomes are analysed and evaluated on a regular and ad hoc basis. • Decisions are made as to which safety-critical problems will be addressed and in which order. Performance • Events in which patients are harmed or nearly harmed are investigated from a safety point of view for the context of care and not for the performance of the individuals involved. • Everyone's evaluation is part of an ongoing cycle of structural redevelopment, process redeployment and outcome review. • The basis for everyone's evaluation is his or her well-being, including the patient's, which contributes to the success of the organisation.

accident: 'An occurrence associated with the operation of an aircraft which takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked … ' [43].

ICAO's definition focuses on the outcomes of the people on board (and the aircraft) – and not on the possibility of errors contributing to causation, whereas the IAEA's focuses on processes, including errors made by the operator. This latter point reinforces current assumptions that 'there is always some identifiable, personal fault that leads to an imperfect outcome' [4]. Such assumptions are contrary to basic tenets of SMS in aviation and other industries, where errors are considered unintentional, cannot be prevented and should not be punished.

Radiation oncology is moving in the direction of greater standardisation of terminology. Several international groups, including the IAEA (SAFFRON), the Radiation Oncology System Incident Reporting System (ROSIS), the Canadian Partnership for Quality Radiotherapy [44] and Radiation Oncology Incident Reporting System (RO-ILS), provide recent examples of attempts at common taxonomy and pooling of event reports, so that other departments can learn from events reported in the databases. These databases can be interrogated using common terms for specific incidents related to a particular work process, treatment technique or a certain type of machinery. This can be particularly useful when departments plan to establish a new technique or technology: it is helpful to look for other departments' reports about these, to reduce the probability of repeating the same event.

SMS requires information from three sources: proactive, concurrent and reactive. Proactive analyses, using FMEA or other methods, look for hazards and hazardous processes that could lead to patients being harmed or nearly harmed (close calls) [45]. For example, detailed FMEA evaluations were conducted twice (in 2006–2007 and 2009) at Johns Hopkins University. These evaluations identified 127 and 159 failure modes, respectively, with systems developed to address these ranked as highest priority [36,46]. Such processes assist organisations to improve staff focus on safety, and emphasising hazard identification and action.

Concurrent evaluations are often labelled as 'prospective' in that they may follow a group of patients undergoing either standard or new treatments. An example is that of real-time audits, providing important direct and indirect patient benefits, addressing issues related to physician prescribing and clinical judgement [37].

Reactive information most often comes from voluntary, confidential reports of incidents and events in which patients were harmed or nearly harmed, and/or events in which personnel encountered problems with equipment [47]. Importantly, although reporting systems can be used in a proactive manner to identify hazards and to help prioritise incidents for investigation, learning and action, there is too heavy a reliance in most medical systems on collating, measuring and benchmarking, and less emphasis on learning [48]. This differs enormously from aviation, which, although still ensuring that reporting takes place, concentrates significantly more effort on prioritisation, investigation and action. As Macrae [48] states: 'reporting constitutes one component of a broad range of conversations and activities focused on safety and risk'. In addition, many medical reporting systems may link with the organisation's managerial structure [48], eliciting fears of retribution, for report content or numbers. Independent safety systems of reporting and review, as in aviation, are preferable.

Having only one source of information, such as proactive, concurrent or reactive, is insufficient for overall safety. For example, in the FMEA study from Johns Hopkins University, '42% of actual errors the incident reporting system identified' were not found by the FMEA process [46].

Many healthcare reporting systems also rely heavily on measuring reporting rates, and benchmarking departments over time and against other departments [49], including one report from one of the authors of this paper (GPD). This is not considered useful in aviation and other safety-focused industries, nor in some areas of health [49]. These rates are artificial and are more related to an organisation's reporting culture and individual interpretations of definitions used to identify reportable incidents. Rates do not monitor safety, but rather monitor reporting [20]. Focusing on maximising event rates also encourages unnecessary reports of minor risk to an organisation (and not patients). This can also lead to overwhelming of the reporting system, with subsequent inability to identify and react to reports warranting greater attention [48,49]. Reductions in 'error rates' could mean real safety improvements or could merely reflect a reduction in reporting, because of changes in attitudes towards reporting, its potential negative repercussions, or even a definition change for types of incident to be reported. Improved error reporting rates represent a 'strong culture of reporting but a poor culture of learning' [48].

Many department-specific systems have been developed that have provided limited learning, focusing more on the quality, collation and benchmarking of incident reports, rather than investigation and action [49]. But having 'information' is also not enough. Information must be analysed, evaluated and used to make system-level changes. One example of structural and procedural system redesign is that of preparing and administering vincristine in and from an intravenous minibag containing a volume too great for intrathecal administration [50]. Structural and procedural changes were also described by Terezakis *et al.* [46]. Another structural example was the institution of a computerised order entry system for chemotherapy prescribing [51]. A systematic review by Kukreti *et al.* [40] also provides answers to important questions such as 'Does a CPOE generate new errors', the type of question that safety-conscious organisations need to ask before and after the implementation of all 'fixes'. An SMS must ensure a mechanism whereby departments and organisations learn from events reported by others, and of any 'fixes' instituted, to save repetition of the same problems.

There is, of course, a cost to the acquisition and use of information: this is where the investment in safety comes in. However, transferring the costs of 'counting errors' to 'investigating harm to patients and instituting fixes' could save money. In addition, to know of system-level hazards and to ignore them is both a moral failing and a potentially legal one, which then would incur expenditures better spent on the improvement side of the ledger.

Summary

Despite some clear differences between aviation and oncology with respect to safety, there are some key lessons that can be taken from the aviation industry to improve safety. There remains a need to ensure that these are instituted systematically throughout healthcare rather than the current sporadic way that safety is managed and reported, often without a centralised mandate.

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